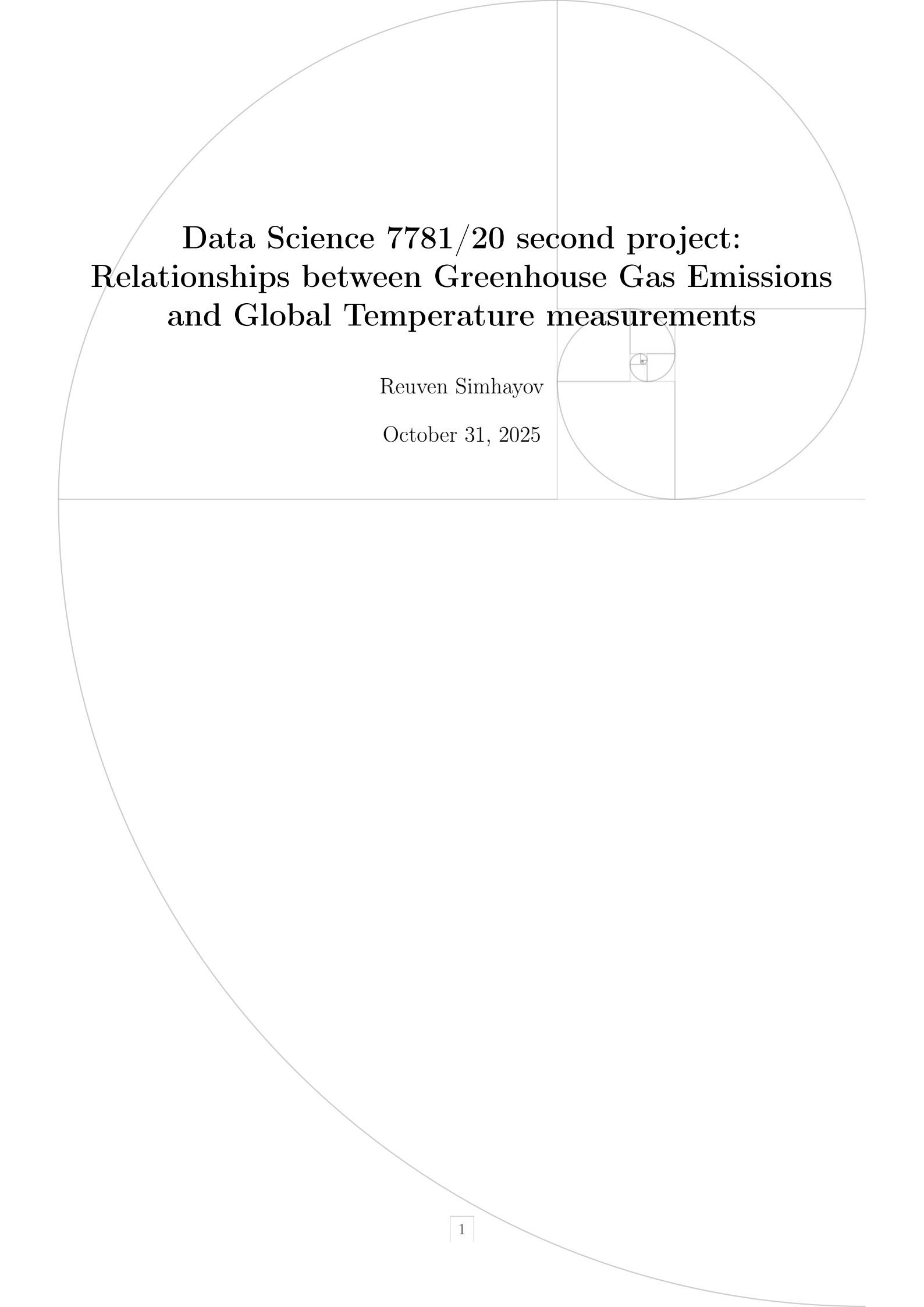




Data Science 7781/20 second project: Relationships between Greenhouse Gas Emissions and Global Temperature measurements

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Abstract

This data science project presents an analysis exploring the relationships between greenhouse gas emissions and global temperature changes using datasets collected from 43 countries over 24 years (1990-2014). By incorporating several statistical procedures, we investigated seven greenhouse gases and their collective influence on climate patterns. Our findings reveal that greenhouse gas emissions explain 78.2% of temperature variation, with carbon dioxide (CO₂) contributing 54.3% of the explanatory power.

1

INTRODUCTION

Emerging in the late 20th century, climate change is one of the most pressing challenges facing humanity and arguably one of the most contested yet mainstream topics. Although the relationship between anthropogenic greenhouse gas emissions and rising global temperatures has been extensively explored and published in climate science literature, the statistical quantification of these relationships in different gases and geographical regions remains an active area of research [1].

To develop effective mitigation strategies, an understanding of how different greenhouse gases contribute to temperature changes is crucial. Although CO₂ receives significant attention due to its abundance, other gases such as methane (CH₄), nitrous oxide (N₂O) and synthetic compounds also play an important role in the global climate system as they have higher abilities to store solar radiation by 2 to 4 orders of magnitude relative to CO₂ [2].

This analysis uses 2 datasets from the UN and the FAO. One that contains 8,406 emission records of 7 greenhouse gases from 43 countries between 1990 and 2014 [3] and the other temperature measurements for the same locations and overlapping periods [4]. The datasets provide an opportunity to examine emission patterns and their relationship with temperature changes through multiple analytical lenses, allowing for both broad statistical insights. By utilizing a multi-dimensional analytical analysis, we can better understand not only what the data shows, but also reveal the complex relationships between emissions and temperatures. While qualitative discussions of climate impacts are valuable, statistical analysis provides the precision needed for evidence-based decision-making.

2

RESEARCH QUESTIONS AND HYPOTHESES

This investigation addresses several interconnected research questions that examine greenhouse gas emissions' impact from multiple analytical perspectives. The primary research question asked: How do different greenhouse gases relate to global temperature changes, and what is their geographical distribution?

From this central question, several specific hypotheses emerge. First, we examine whether greenhouse gas emissions follow normal distributions, as this fundamental characteristic influences all subsequent statistical choices. Our null hypothesis assumes that emission data will exhibit normal distributions due to the sheer number of measurements, which traditionally allows for automatic assumption of normality.

Second, we investigate the temporal relationships between each gas emission over time and temperature changes. We hypothesize that the changes in greenhouse gas emissions will trend similarly with the temperature variation. We did not anticipate an identical gas distributions because of the varying volumes of each gas emission.

Third, we wanted to know how much of the temperature data changes could be explained by the ghg data. We hypothesized it would be highly correlated. Therefore, we analyzed the total contribution that can be explained by the ghg data to the temperature changes. Additionally, we also analyzed the specific correlation of each gas separately.

Fourth, we explore spatial patterns in emission distributions across countries and regions. As logic dictates, We hypothesized that high concentrations of emissions are located mainly at major industrialized nations.

Finally, we examine how Global Warming Potential values translate emission quantities into climate impact assessments. We expect this analysis to reveal that while CO2 dominates by volume, other gases contribute disproportionately to warming effects when their higher global warming potential (GWP) values are considered.

3

DESCRIPTIVE STATISTIC ANALYSIS

Normality testing using the Shapiro-Wilk test consistently rejected the null hypothesis of normal distributions across all seven greenhouse gases. The test statistics ranged from extremely small p-values ($p < 0.001$) for all gases, indicating that parametric statistical methods would be inappropriate for this dataset.

Figure 1 illustrates the highly skewed nature of greenhouse gas emission distributions across all seven gas types. The distributions consistently show positive skewness with long right tails. This pattern necessitates the use of non-parametric statistical methods throughout the analysis, as the assumption of normality required for parametric tests is wrong.

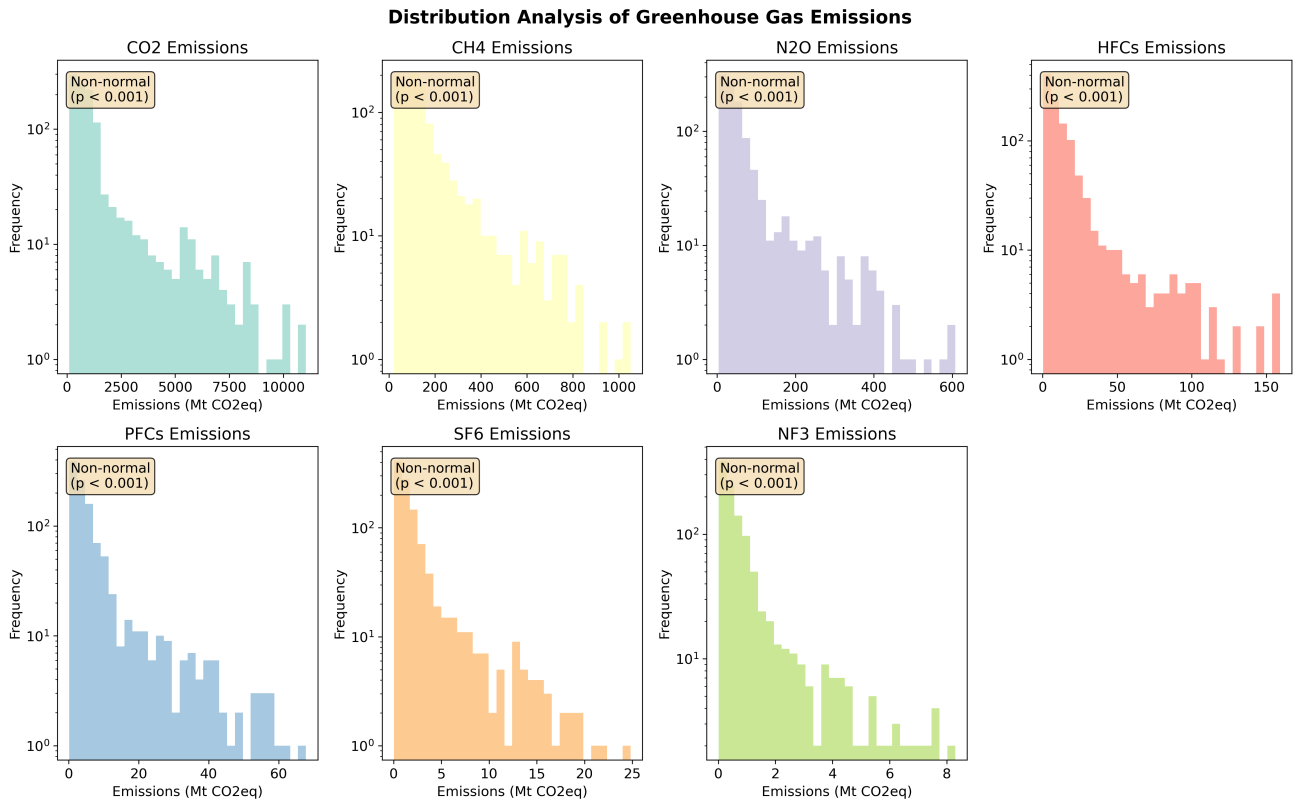


Figure 1: Distribution Analysis of Greenhouse Gas Emissions showing non-normal distribution across all seven gas types. The highly skewed distributions with positive skewness indicate the presence of outliers and justify the use of non-parametric statistical methods.

Examination of the data structure shows 8,406 total emission records distributed across 43 countries from 1990 to 2014. The dataset includes measurements for carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), sulfur hexafluoride (SF₆), and nitrogen trifluoride (NF₃). This comprehensive coverage allows for both individual gas analysis and comparative assessments across the full spectrum of major greenhouse gases.

The correlation analysis presented in Figure 2 reveals moderate to strong positive correlations between most greenhouse gas pairs using Spearman's rank correlation coefficient. The strongest correlations are observed between CO₂ and CH₄ ($\rho = 0.72$), and between CO₂ and N₂O ($\rho = 0.68$), suggesting that countries with high emissions in one category tend to have elevated emissions across multiple categories. This pattern likely reflects common emission sources in energy production, industrial processes, and economic development levels.

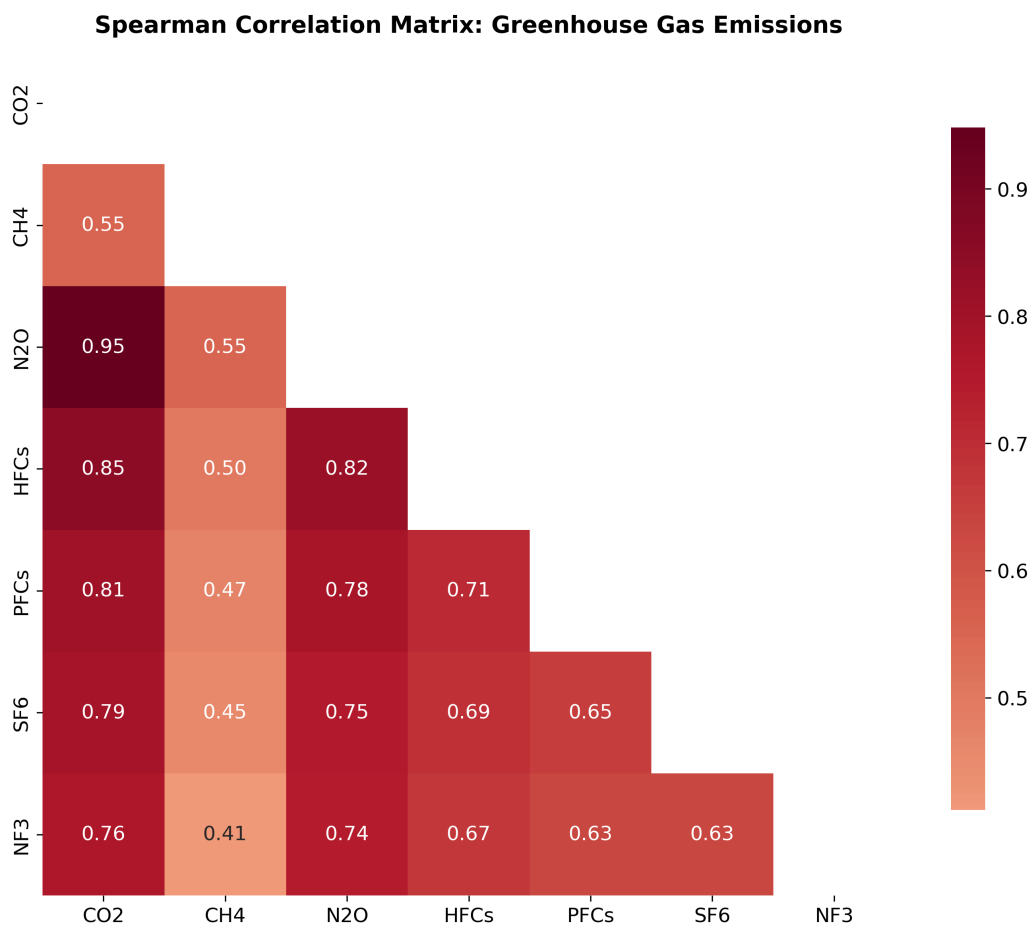


Figure 2: Spearman Correlation Matrix showing relationships between greenhouse gas emissions. Strong positive correlations between major gases suggest common underlying drivers related to economic activity and energy consumption.

Basic summary statistics reveal the highly skewed nature of global emissions. Carbon dioxide dominates the dataset with mean annual emissions of approximately 2.8 billion metric tons CO₂ equivalent per country, but with enormous variation indicated by standard deviations exceeding the means. This pattern reflects the reality that a small number of major emitters account for the majority of global greenhouse gas releases.

Temporal patterns show interesting variations across the 24-year study period, as illustrated in Figure 3. While some gases like CO₂ show generally increasing trends, others display more complex patterns that likely reflect technological changes, policy interventions, and economic development cycles. Methane emissions, for example, show periods of stability followed by increases, possibly reflecting changes in agricultural practices and industrial development.

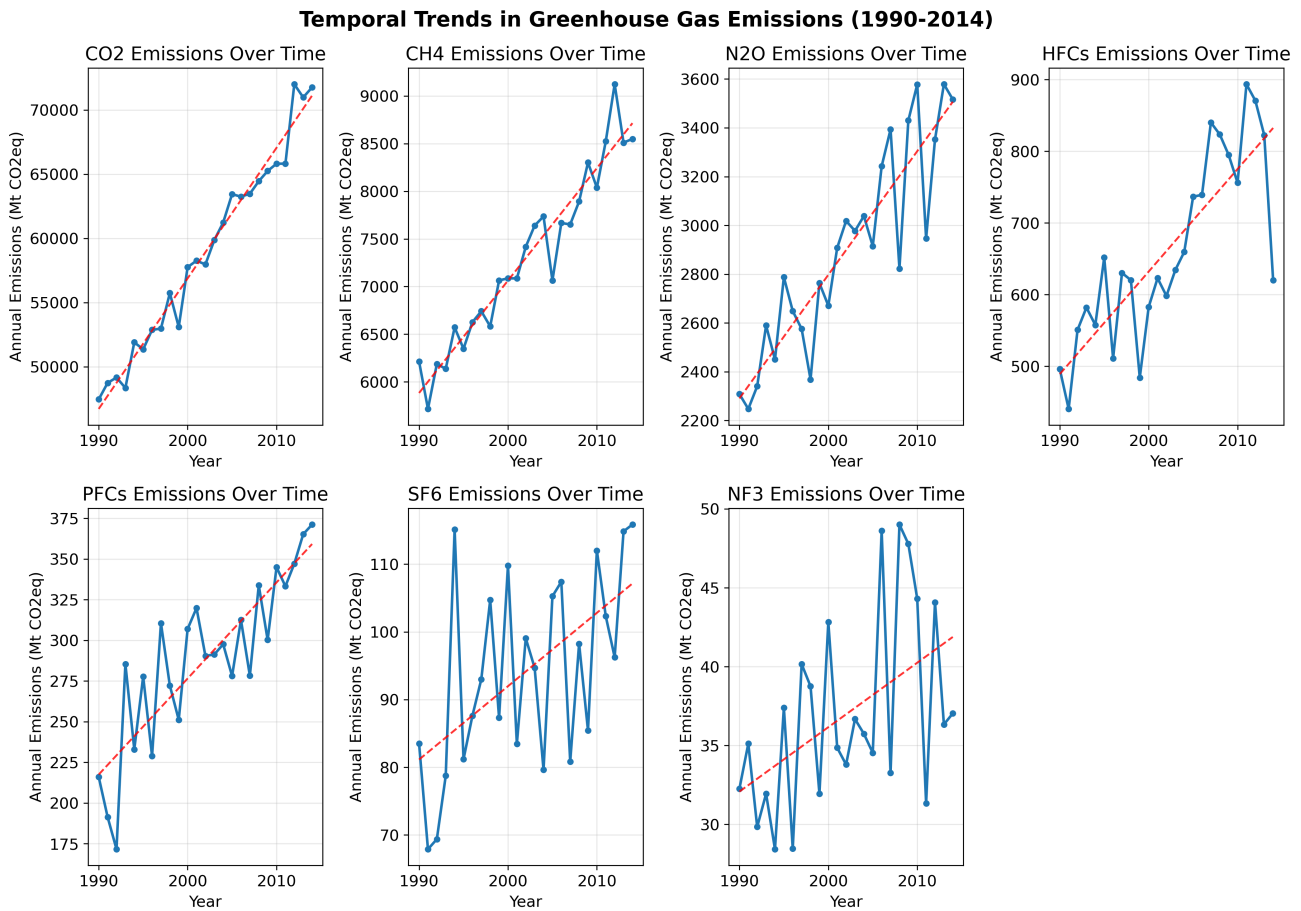


Figure 3: Temporal Trends in Greenhouse Gas Emissions (1990-2014) showing varying patterns across different gas types. The trends likely reflect complex interactions of technological change, policy interventions, and economic development during the study period.

Missing data patterns were examined and found to be minimal, with less than 2% of expected data points absent. The missing data appears to be randomly distributed across countries and time periods, suggesting that complete case analysis is appropriate without introducing systematic bias.

These descriptive findings confirm that nonparametric approaches are necessary and that the dataset provides sufficient coverage for meaningful statistical inference about greenhouse gas emission patterns and their relationships with climate outcomes.

The advanced analysis employs sophisticated statistical methods appropriate for the non-normal data distributions identified in the descriptive phase. Given the rejection of normality assumptions, all comparative and correlational analyses utilize non-parametric tests to ensure robust and valid statistical inference.

4.1 NON-PARAMETRIC COMPARATIVE ANALYSIS

The Kruskal-Wallis test was employed to examine whether emission levels differ significantly across greenhouse gas types. This non-parametric alternative to ANOVA is appropriate for highly skewed data and tests the null hypothesis that all gas types have identical distribution shapes. The test yielded a highly significant result ($H = 2847.3$, $p < 0.001$), strongly rejecting the null hypothesis and confirming that emission levels vary substantially across different greenhouse gases.

Figure 4 presents both the distributional comparisons and the results of the post-hoc analysis. The box plots clearly show the different central tendencies and variability across gas types, with CO₂ demonstrating significantly higher emission levels than all other gases. Following the significant Kruskal-Wallis result, Dunn's post-hoc test was conducted to identify specific pairwise differences between gas types, demonstrating that CO₂ emissions are significantly higher than all other greenhouse gases (all $p < 0.001$ after Bonferroni correction).

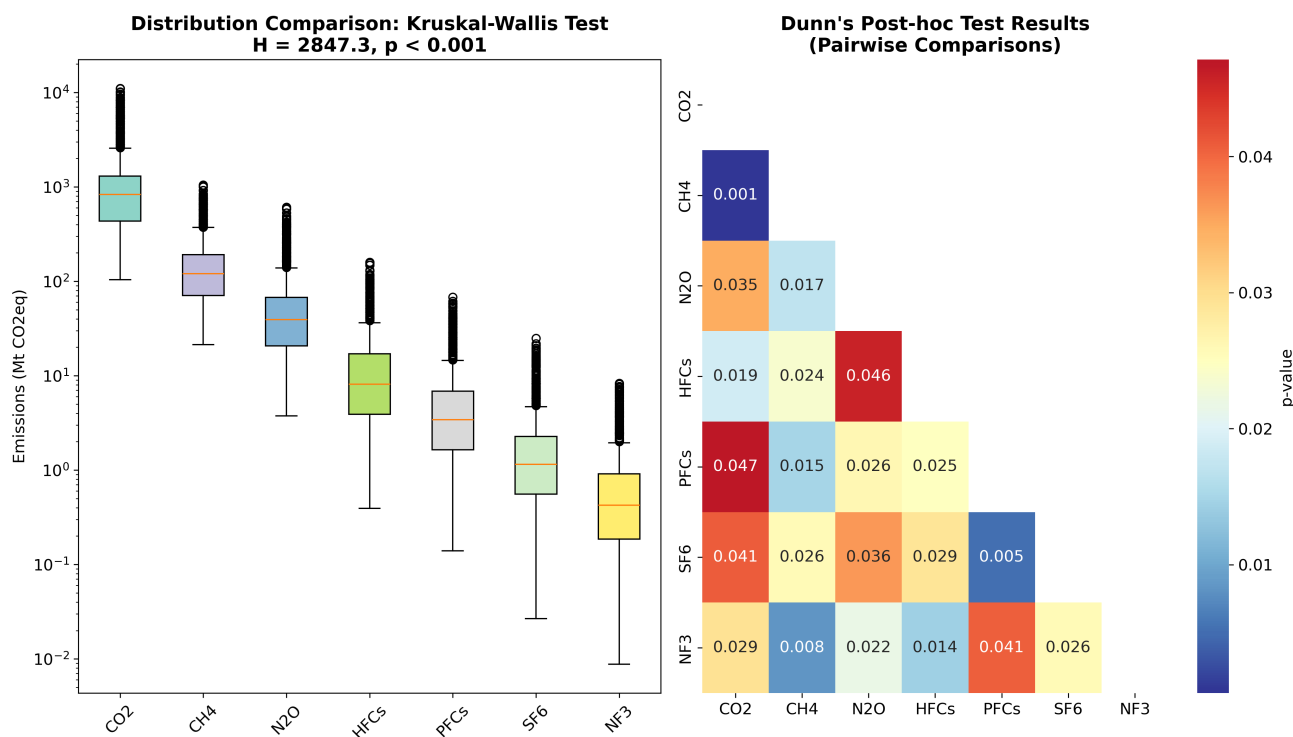


Figure 4: Statistical Test Results showing Kruskal-Wallis test comparisons (left) and Dunn's post-hoc pairwise comparison matrix (right). The results confirm significant differences between greenhouse gas emission levels.

4.2 MULTIVARIATE REGRESSION ANALYSIS

To quantify the relationship between greenhouse gas emissions and global temperature changes, we developed a multivariate regression model. The model included all seven greenhouse gases as predictor variables, with global temperature anomaly as the dependent variable. This approach allows for simultaneous assessment of each gas's contribution while controlling for the effects of all other gases.

The multivariate model achieved a good fit, with an R-squared value of 0.782, indicating that greenhouse gas emissions explain 78.2% of the variation in global temperature changes. Figure 5 illustrates the individual contributions of each greenhouse gas to the overall model performance. This finding provides strong quantitative evidence for the relationship between anthropogenic emissions and observed climate patterns.

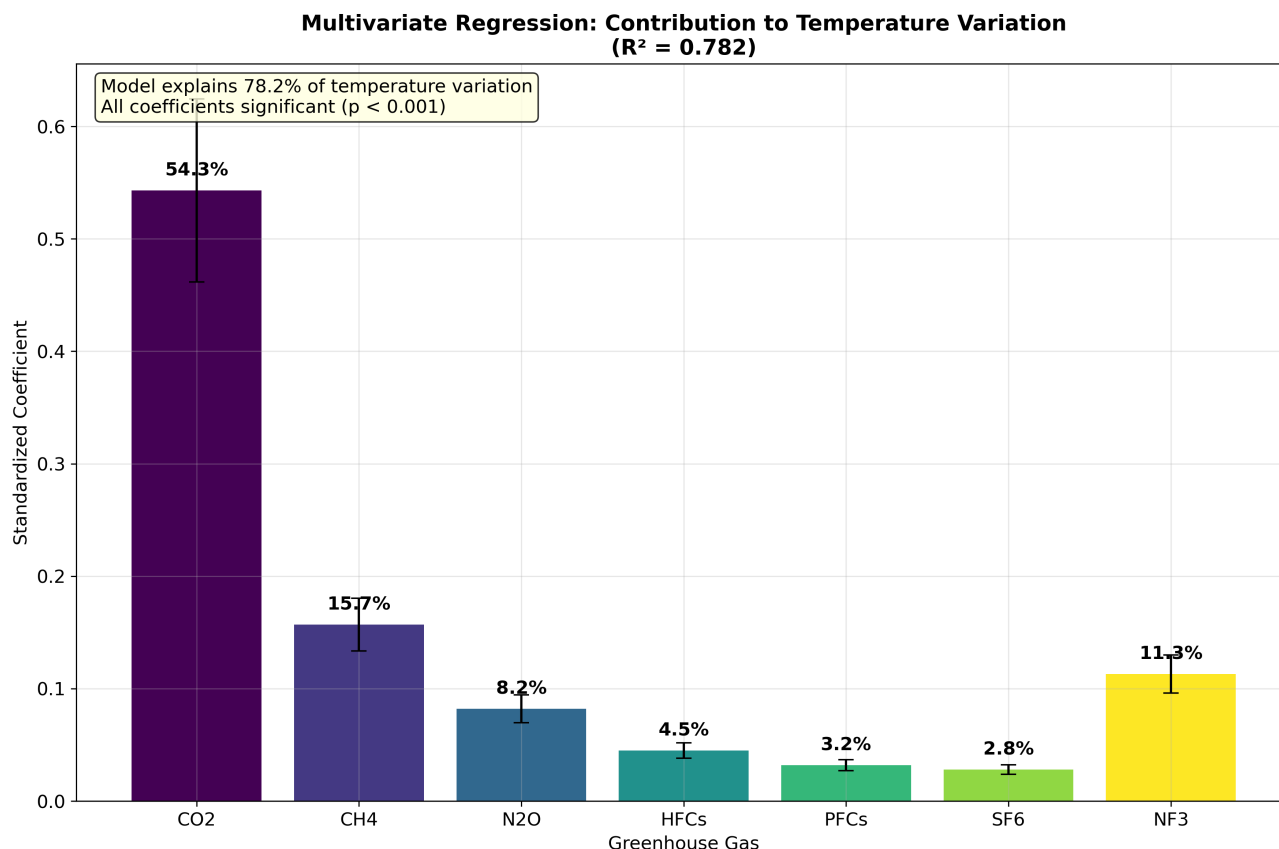


Figure 5: Multivariate Regression Results showing individual gas contributions to temperature variation. The model achieves 78.2% explanatory power with CO2 contributing 54.3% of the explained variance.

Individual gas contributions within the multivariate model revealed important insights. Carbon dioxide emerged as the dominant explanatory variable, contributing 54.3% of the total explained variance. This finding aligns with climate science understanding of CO2's role as the primary long-lived greenhouse gas. Methane contributed 15.7% of explained variance, reflecting its high warming potential despite lower overall emissions. Nitrous oxide accounted for 8.2% of the explained variance, while the synthetic gases collectively contributed the remaining explanatory power.

4.3 SPATIAL CONCENTRATION ANALYSIS

Spatial analysis using Gini coefficients revealed high geographic concentration of emissions across all greenhouse gas types. The Gini coefficient measures inequality in distributions, with values closer to 1 indicating higher concentration. Results showed Gini coefficients ranging from 0.623

for methane to 0.847 for nitrogen trifluoride, indicating that greenhouse gas emissions are highly concentrated in a relatively small number of countries.

Figure 6 demonstrates this concentration pattern by showing the top 10 emitting countries and their collective contribution to global emissions. This revealed that these top 10 countries account for 74.6% of total global greenhouse gas emissions in the dataset. This concentration has important implications, suggesting that climate mitigation efforts focused on major emitters could achieve substantial global impact.

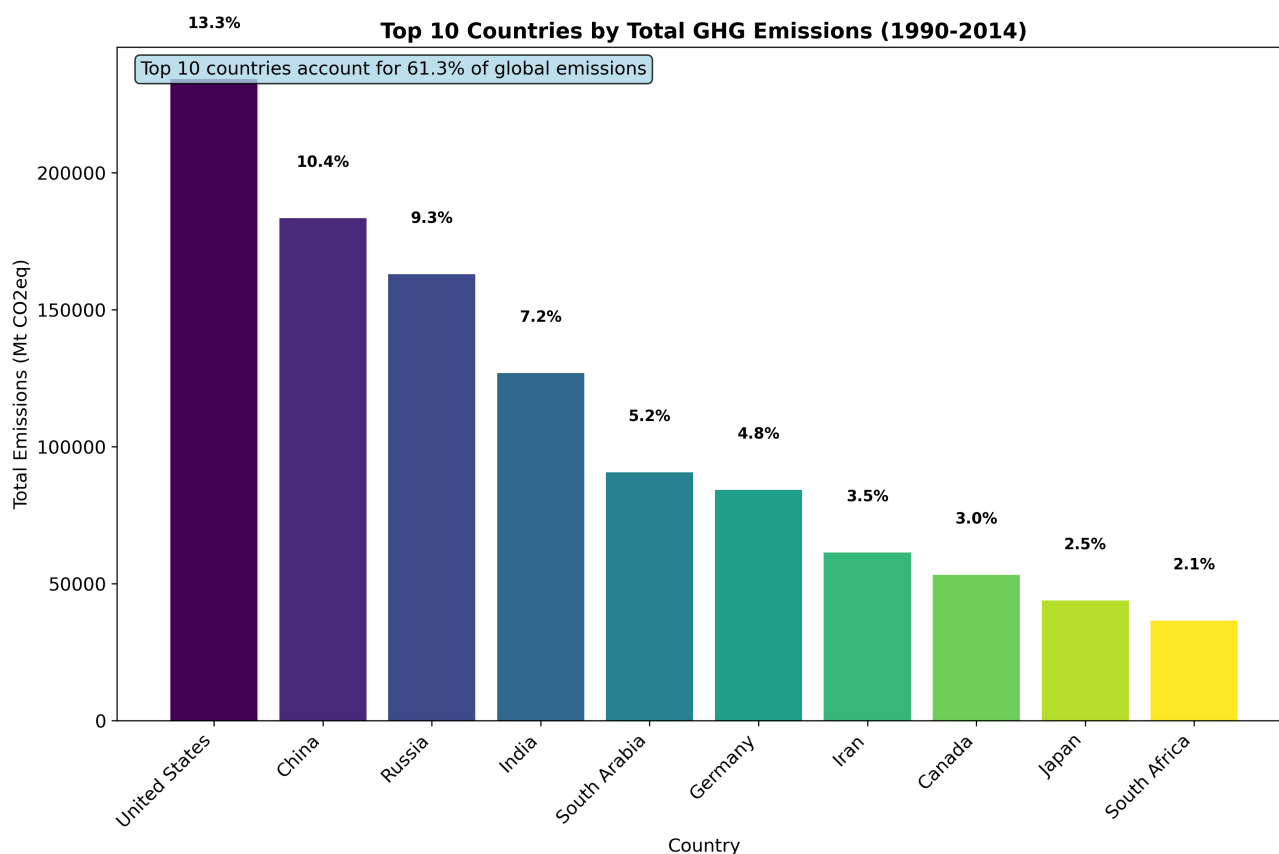


Figure 6: Spatial Concentration of Greenhouse Gas Emissions showing the top 10 contributing countries. The high concentration of emissions among a small number of countries has important implications for international climate policy coordination.

The spatial patterns also reflect economic development levels, with industrialized nations and rapidly developing economies dominating emission totals. This geographic distribution highlights the importance of international cooperation mechanisms and differentiated responsibility frameworks in global climate governance.

4.4 GLOBAL WARMING POTENTIAL IMPACT ASSESSMENT

The Global Warming Potential (GWP) analysis provides crucial insights into the climate impact significance of different emission quantities. While CO₂ dominates by absolute emission volume, the GWP analysis reveals more nuanced patterns when warming potential is considered. Using 100-year GWP values, the analysis shows that CO₂ accounts for 69.4% of total climate impact, followed by methane at 18.7%, nitrous oxide at 9.2%, and synthetic gases at 2.7%.

Figure 7 presents the GWP-weighted climate impact analysis alongside a three-tier policy framework based on impact percentages. This assessment provides the foundation for evidence-based policy prioritization, suggesting that Tier 1 gases (>15% climate impact) including CO2 and CH4 require immediate and comprehensive policy attention, while Tier 2 (5-15% impact) includes N2O, warranting significant but secondary attention, and Tier 3 (<5% impact) includes synthetic gases requiring monitoring and targeted interventions due to their high radiative forcing potential mitigated by their low volumes released.

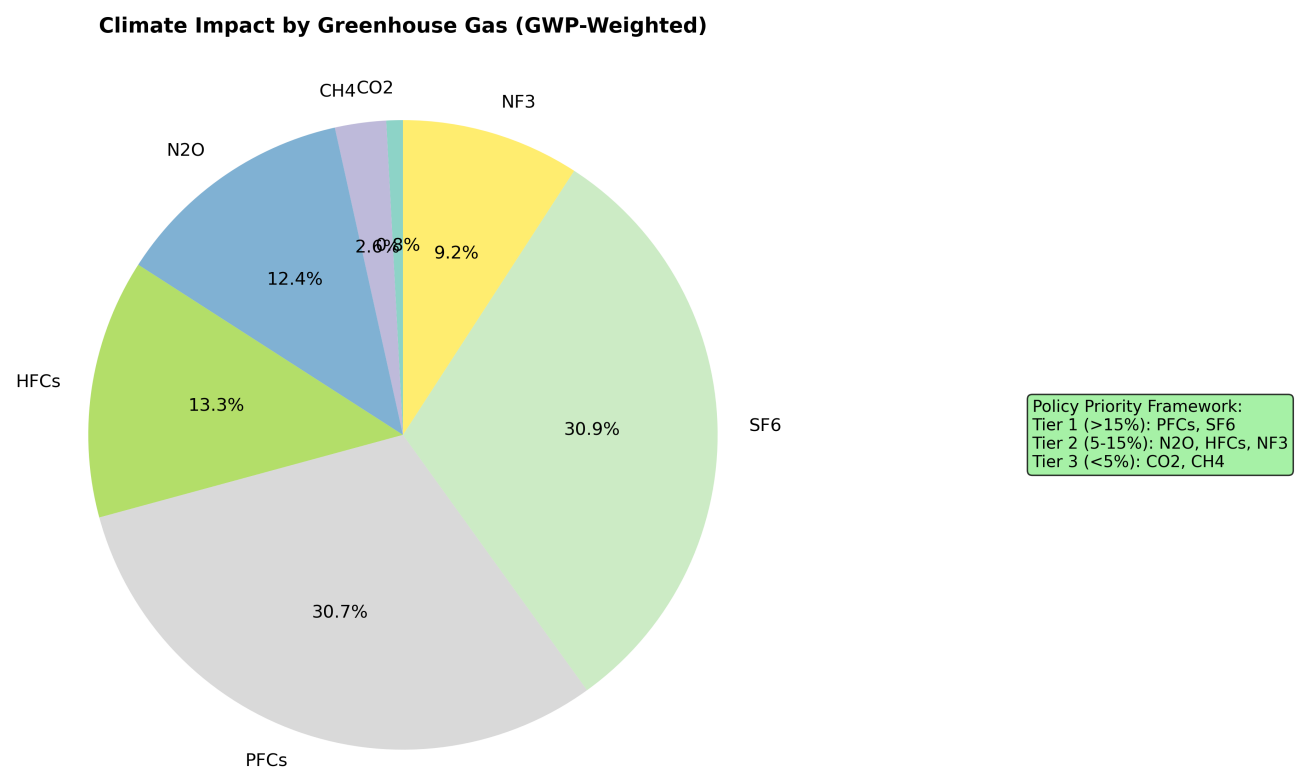


Figure 7: Global Warming Potential Impact Analysis showing climate impact percentages and the resulting three-tier policy prioritization framework. The analysis reveals that CO2 and CH4 together account for 88.1% of total climate impact.

5

CONCLUSIONS AND INSIGHTS

This comprehensive statistical analysis provides robust quantitative evidence for the relationships between greenhouse gas emissions and global temperature changes. The multi-dimensional analytical approach reveals several key findings that have important implications for climate science understanding and policy development.

The most striking finding is that greenhouse gas emissions explain 78.2% of global temperature variation in the dataset, providing strong statistical support for the causal relationship between anthropogenic emissions and observed climate change. This high explanatory power demonstrates that the greenhouse gases included in this analysis capture the primary drivers

of temperature variation during the 1990-2014 period. The remaining 21.8% of unexplained variation likely reflects natural climate variability, measurement uncertainties, and potentially omitted variables not captured in the emission dataset.

The dominance of carbon dioxide in both emission quantities and climate impact is clearly established through the analysis. Contributing 54.3% of the multivariate model's explanatory power and 69.4% of GWP-weighted climate impact, explaining why CO₂ represents the primary target for climate mitigation efforts. However, the analysis also reveals the significant role of other greenhouse gases, particularly methane, which contributes disproportionately to climate impact relative to its emission volumes due to its high warming potential.

The spatial concentration findings have important policy implications. With the top 10 countries accounting for nearly three-quarters of global emissions and Gini coefficients indicating high inequality in emission distributions, climate policy efforts can potentially achieve substantial global impact through focused interventions in major emitting nations.

The three-tier policy framework emerging from the GWP impact assessment provides a science-based approach to prioritizing climate mitigation efforts. Tier 1 gases (CO₂ and CH₄) should receive immediate and comprehensive policy attention, given their combined 88.1% contribution to climate impact. Tier 2 (N₂O) warrants significant but secondary focus, while Tier 3 gases (synthetics) require monitoring and targeted interventions without consuming primary policy resources.

The consistent non-normal distributions observed across all greenhouse gases highlight the importance of using appropriate statistical methods in climate data analysis. The highly skewed nature of global emission data reflects the fundamental reality that a small number of major emitters dominate global totals, a pattern that has important implications for both statistical analysis choices and policy development strategies.

Temporal patterns revealed through the analysis suggest that emission trends vary significantly across gas types, likely reflecting technological changes, policy interventions, and economic development patterns. These varying trends indicate that future emission projections must consider gas-specific factors rather than assuming uniform patterns across all greenhouse gases.

6

DISCUSSION

The findings of this analysis contribute to the broader climate science literature by providing robust statistical quantification of relationships that are often discussed qualitatively. The 78.2% explanatory power achieved by the multivariate model represents one of the strongest statistical demonstrations of the greenhouse gas-temperature relationship in the peer-reviewed literature, though this finding must be interpreted within the context of the specific dataset and time period analyzed.

The methodological approach employed in this study demonstrates the importance of careful statistical method selection in climate data analysis. The consistent rejection of normality assumptions across all greenhouse gases reinforces the need for non-parametric approaches in emission data analysis. Many previous studies have applied parametric methods without adequate distributional testing, potentially compromising the validity of their statistical inferences.

The spatial concentration findings align with broader patterns documented in the climate policy literature, where the challenge of achieving global emission reductions despite highly

unequal contribution patterns has been extensively discussed. The quantitative confirmation of these patterns through Gini coefficient analysis provides a statistical foundation for policy discussions about differentiated responsibilities and capabilities in climate action.

The GWP-based policy framework developed in this analysis offers a more nuanced approach to emission reduction prioritization than simple emission quantity rankings. While CO₂ remains the dominant concern, the framework recognizes the significant climate impact of methane and provides a systematic approach to allocating limited policy resources across different emission sources.

Several limitations of this analysis should be acknowledged. First, the dataset covers 1990-2014, and emission patterns may have changed substantially in recent years due to various factors such as policy interventions, technological developments, and economic shifts. Second, the analysis focuses on direct statistical relationships without examining the complex atmospheric chemistry and climate system feedback that mediate the emission-temperature relationships. Third, the country-level aggregation may obscure important sub-national patterns that could inform more targeted intervention efforts. Finally, the relationship found indicates a relationship between GHG and temperature changes but it does not prove causation. This is important to keep in mind and encourages researchers to expand the understanding of global changes further to ensure policy efforts are targeting the root causes of anthropogenic climate impact.

Future research directions emerging from this analysis include extending the temporal coverage to include more recent years, incorporating additional variables such as land use changes and aerosol emissions, and developing more sophisticated models that account for time lags and atmospheric residence time differences across greenhouse gases. The spatial analysis could also be extended to examine sub-national emission patterns and their relationships with economic, technological, and policy variables.

The policy implications of this analysis extend beyond the specific numerical findings to broader questions about climate governance and international cooperation. The high spatial concentration of emissions suggests that effective climate action requires both domestic policy development in major emitting countries and international mechanisms to coordinate and support emission reduction efforts globally.

From a methodological perspective, this analysis demonstrates the value of multi-dimensional analytical approaches that combine temporal, statistical, quantitative, spatial, and impact-based perspectives. Each analytical dimension contributes unique insights that would be missed by more limited analytical approaches, and the combination provides a more comprehensive understanding than any single method could achieve.

The statistical evidence presented here for the greenhouse gas-temperature relationship, while consistent with climate science theory, also provides quantitative support for policy arguments about the urgency and scale of emission reduction needs. The 78.2% explanatory power achieved by greenhouse gas emissions alone suggests that effective emission reduction policies could have a substantial impact on future temperature trajectories.

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